Identity Services Lab 1

Local User Authentication and Secure SSH

Cisco Packet Tracer Evidence Portfolio

Project Objective

Build and secure a small Cisco network by configuring local administrator identities, encrypted credentials, SSH version 2, console and VTY authentication, management addressing, security banners, verification tests, and saved startup configurations.

Skills Demonstrated

Identity and access management | Cisco IOS | Secure Shell (SSH) | IPv4 networking | Local authentication | Network troubleshooting | Configuration verification | Secure device administration

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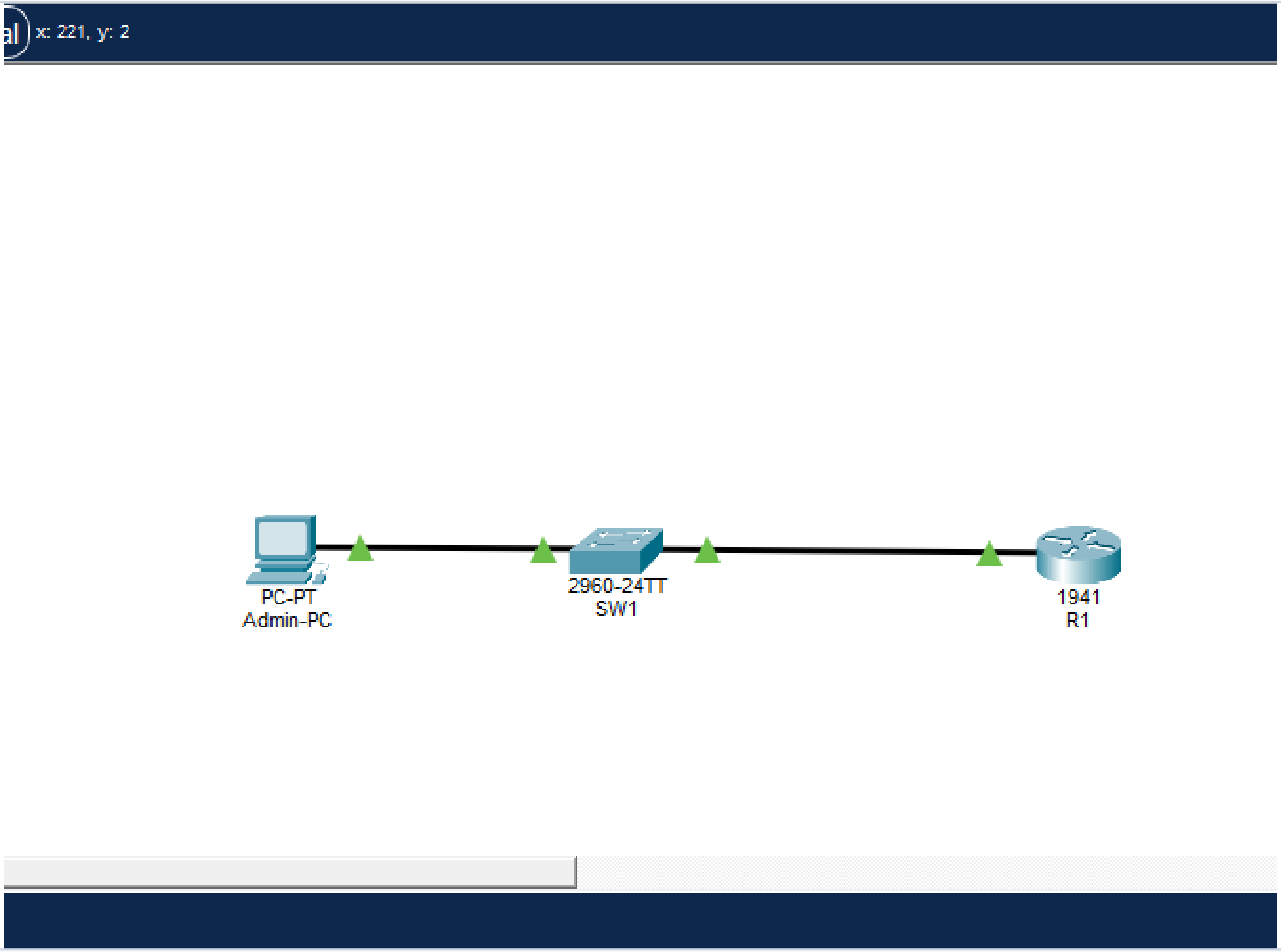
1. Network Topology



Cisco Packet Tracer topology containing an administrator workstation, a Cisco 2960 switch, and a Cisco 1941 router with all links operational.

Topology design | Device naming | Cabling | Interface status verification

2. Connectivity Test to R1

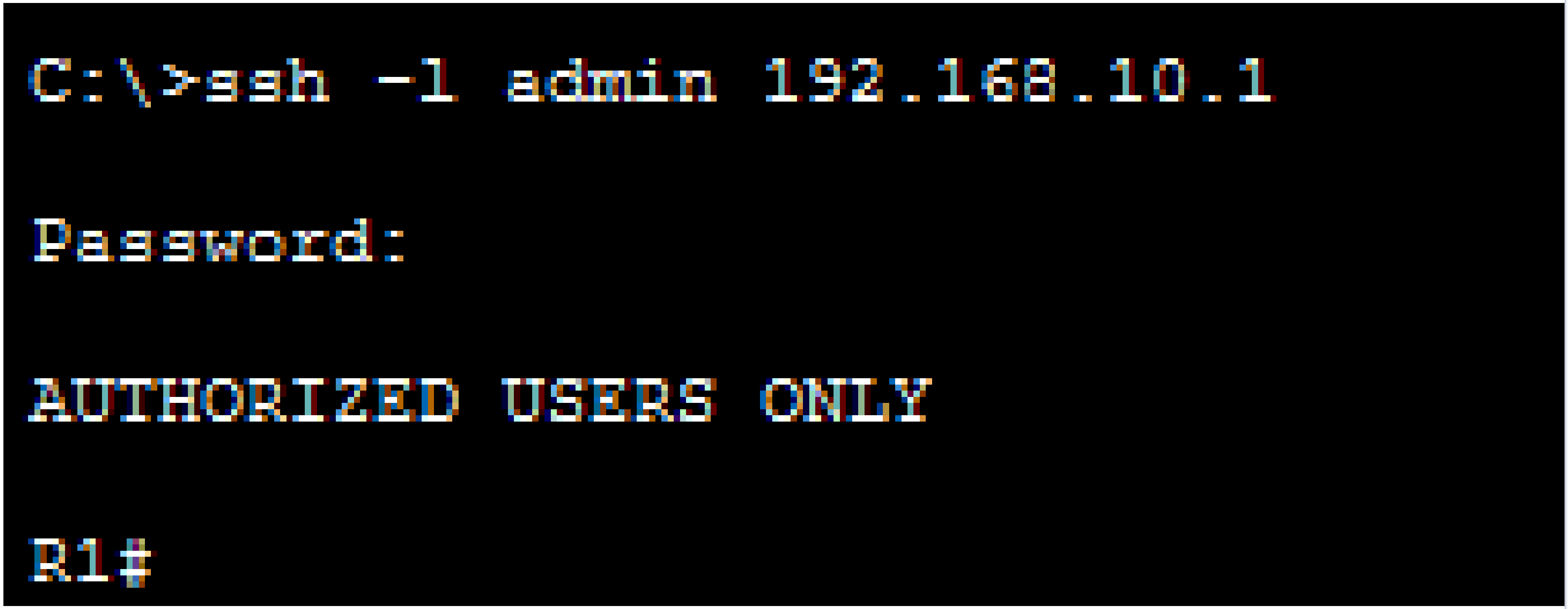
```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Successful ICMP replies confirm that Admin-PC can reach the router at 192.168.10.1 with zero packet loss.

IPv4 addressing | Default gateway configuration | Ping testing | Troubleshooting

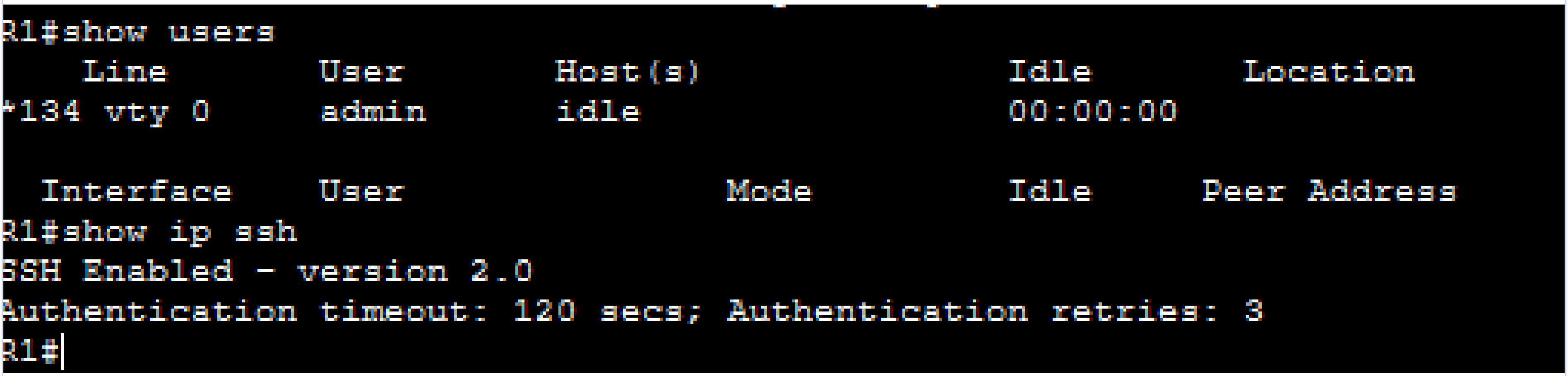
3. Successful Local User SSH Authentication



Admin-PC connects securely to R1 through SSH using a locally configured administrator account and receives the authorized-use banner.

Local identities | SSH version 2 | Privilege level 15 | Secure remote administration

4. Active SSH Session and Version Verification



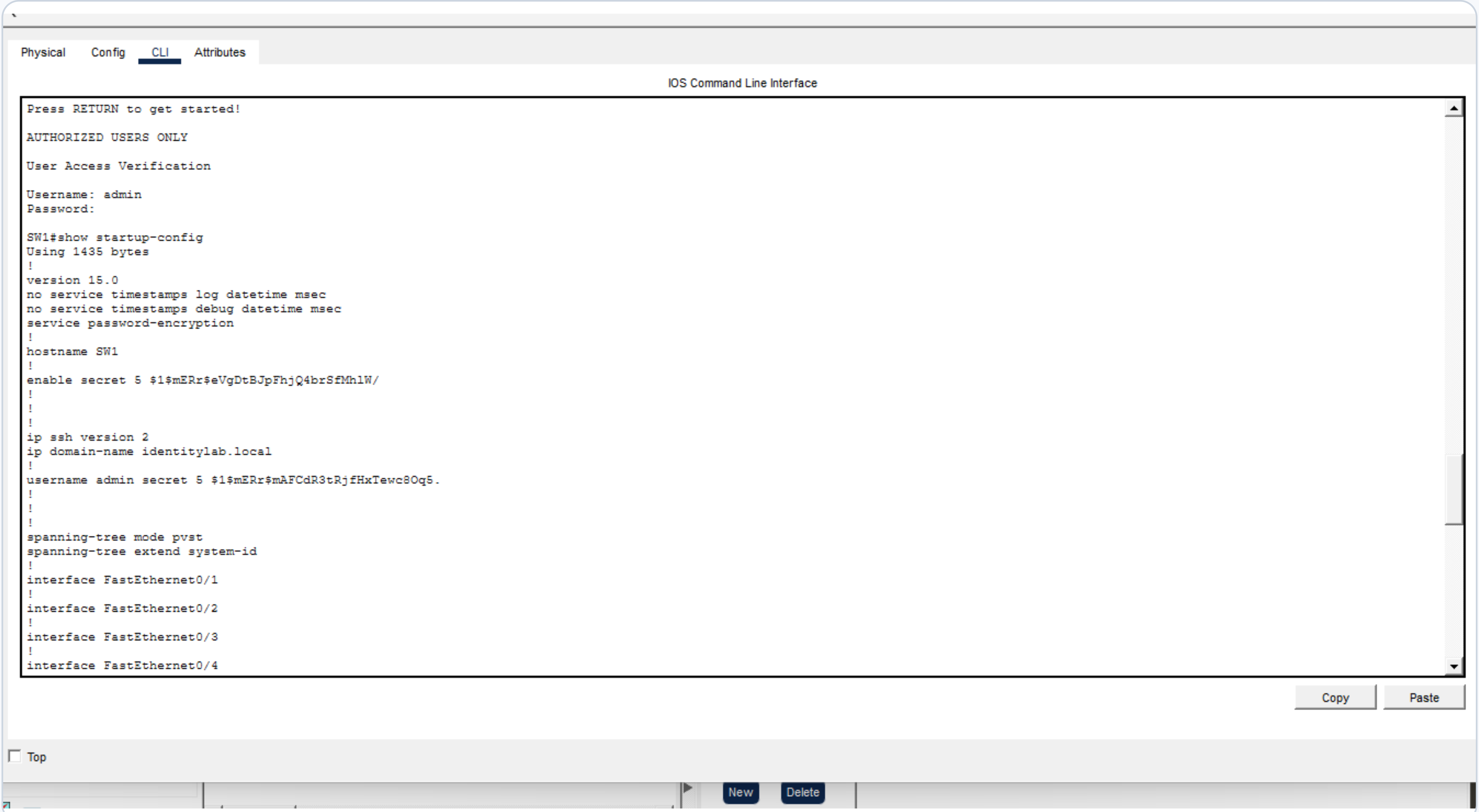
```
R1#show users
    Line       User       Host(s)              Idle       Location
*134 vty 0     admin      idle                 00:00:00

  Interface    User               Mode         Idle     Peer Address
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#
```

The show users and show ip ssh commands verify the active administrator session and confirm that SSH version 2 is enabled.

Session verification | Cisco IOS show commands | Authentication validation | SSH hardening

5. Saved SW1 Identity and SSH Configuration



The startup configuration confirms the switch hostname, encrypted enable secret, local administrator account, domain name, and SSH version 2 settings.

Configuration persistence | Encrypted credentials | Switch management | Identity services